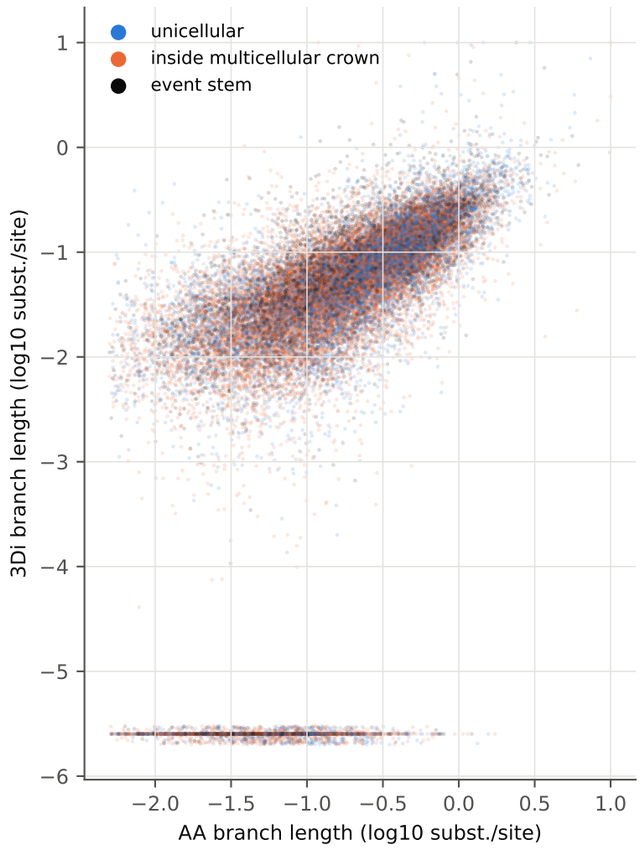
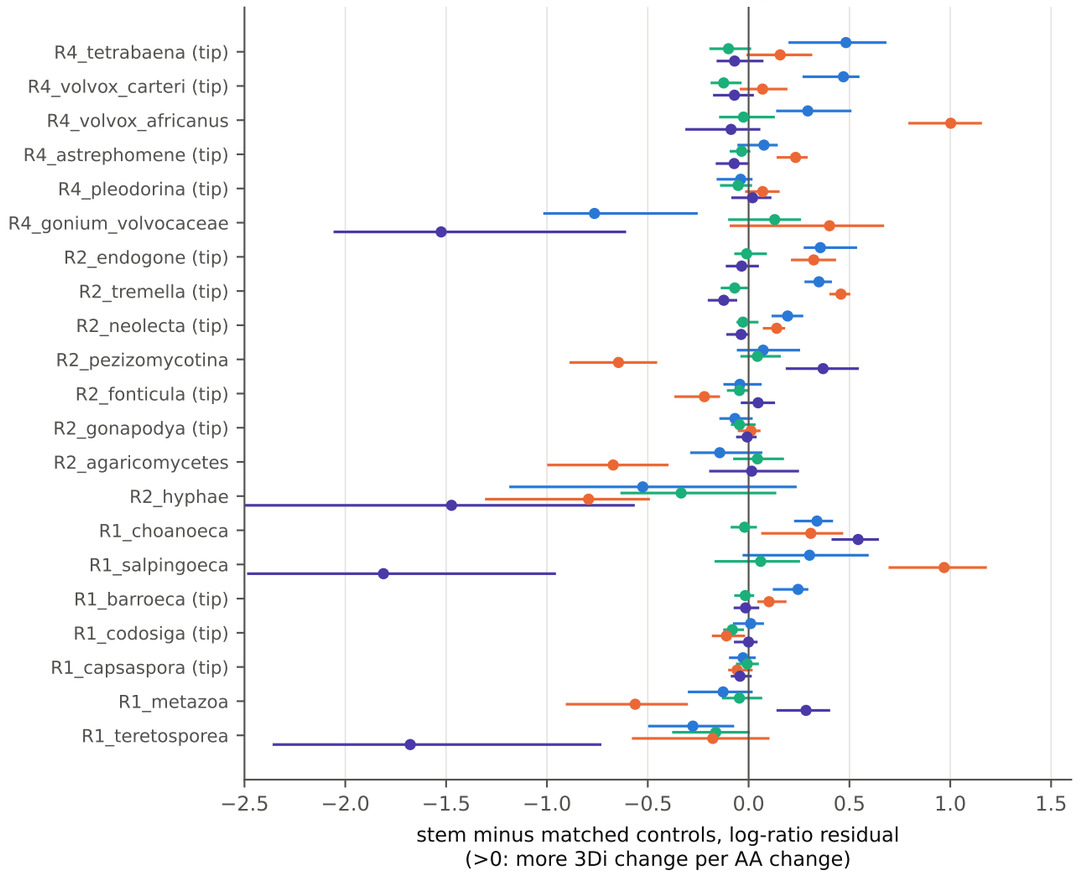
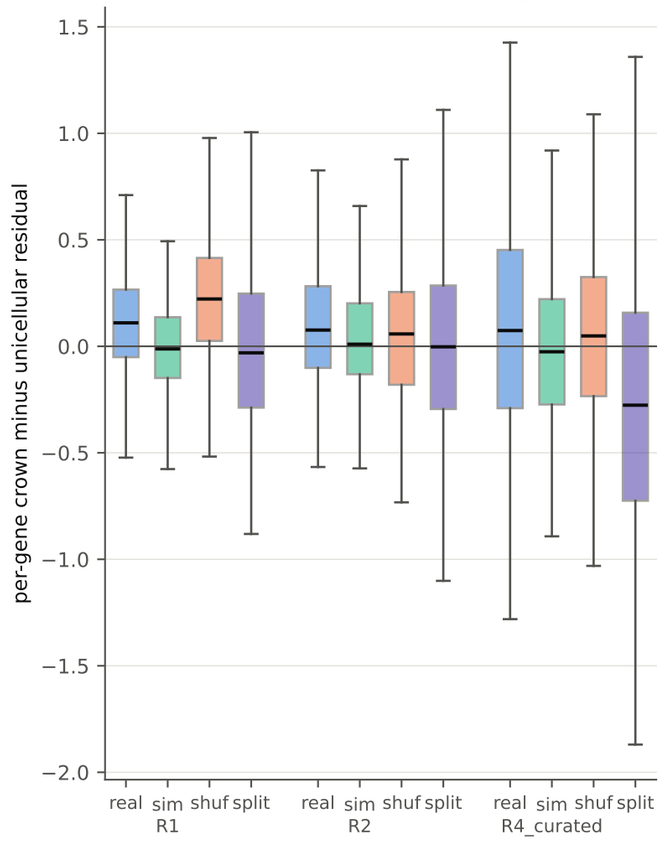

A 3Di tracks AA length (3Di = 0 drawn at 1e-6)
unicellular
inside multicellular crown
event stem
3Di branch length (log10 subst./site)
AA branch length (log10 subst./site)


B Event stems vs matched controls (95% gene-bootstrap CI)
R4_tetrabaena (tip)
R4_volvox_carteri (tip)
R4_volvox_africanus
R4_astrephomene (tip)
R4_pleodorina (tip)
R4_gonium_volvocaceae
R2_endogone (tip)
R2_tremella (tip)
R2_neolecta (tip)
R2_pezizomycotina
R2_fonticula (tip)
R2_gonapodya (tip)
R2_agaricomycetes
R2_hyphae
R1_choanoeca
R1_salpingoeca
R1_barroeca (tip)
R1_codosiga (tip)
R1_capsaspora (tip)
R1_metazoa
R1_teretosporea
stem minus matched controls, log-ratio residual
(>0: more 3Di change per AA change)
3Di / AA (real)
constant-ratio simulation (null)
shuffled 3Di / AA (null)
AA half / AA half (null)


C Crown vs unicellular branches (genes)
per-gene crown minus unicellular residual
real
sim
shuf
split
R1
R2
R4_curated